

HITESH GUPTA

Bhopal, Madhya Pradesh, India

+91-9610788676 [✉ hiteshgupta5151@gmail.com](mailto:hiteshgupta5151@gmail.com) [in LinkedIn](#) [GitHub](#)

About Me

AI and ML student with hands-on experience building intelligent systems across computer vision, deep learning, and predictive modelling. Strong foundation in Python, NumPy, pandas, Scikit-learn, and OpenCV, with practical experience in object detection, image processing, and GANs. Committed to applying cutting-edge ML techniques to solve real-world challenges.

Education

Vellore Institute Of Technology <i>B.Tech in Computer Science and Engineering (AI & ML)</i>	Sep. 2023 – May 2027 <i>CGPA: 8.64</i>
DS Science Academy <i>Class XII</i>	July 2021 – April 2022 <i>Percentage: 89.80%</i>

Experience

Edunet Foundation (In association with Microsoft and AICTE) <i>AI and Machine Learning Intern</i>	June 2025 – July 2025 <i>Remote</i>
- Gained hands-on experience with core ML concepts including supervised and unsupervised learning, classification, regression, and unlabeled data handling through practical labs on Microsoft Learn and Edunet LMS. - Explored Generative AI and Computer Vision on Microsoft Azure, working with Azure Custom Vision for image classification and real-world Gen AI applications through practical demos. - Developed “AI-Based Medical Image Enhancement for Diagnostics Using GANs” – a GAN-based system to enhance low-resolution medical images, improving diagnostic accuracy in resource-constrained healthcare settings.	

Projects

Human Sustainability Score Prediction Model <i>Python, Scikit-learn, pandas, NumPy, Matplotlib</i>
- Designed an AI-driven modular pipeline with four components — Electricity, Transport, Recycling, and Tech — to predict individual sustainability scores by converting lifestyle inputs into standardized CO₂ equivalents using logistic scaling functions. - Trained and evaluated six ML models including Linear Regression, Random Forest, and Gradient Boosting, with Linear Regression achieving the best performance at R² = 0.68 and MAE = 0.37. - Visualized model performance through MSE comparisons, residual error distributions, and score distribution graphs to support data-driven environmental policymaking.

Lane Detection and Driver Assistance System <i>Python, NumPy, OpenCV, Ultralytics YOLO11</i>
- Built an end-to-end computer vision pipeline for lane segmentation and steering angle estimation, training a YOLO11n-seg model on the Indian Driving Dataset (IDD) with 6,993 annotated road images. - Engineered a mask-to-YOLO annotation pipeline using OpenCV contour detection to convert segmentation masks into polygon labels, enabling an 80/20 train-validation split for supervised training. - Designed a geometric steering angle algorithm ($\pm 30^\circ$ range) from lane-centroid offset, and implemented inference scripts for both dataset and external images with visual overlays for interpretability.

Smart Retail Checkout System <i>Python, YOLOv8, FastAPI, React.js, PyTorch, Ultralytics</i>
- Built an end-to-end AI-powered retail billing system by training a YOLOv8s model on the IIT Patna Grocery Dataset (5,983 images, 1,816 product classes) for 100 epochs, achieving mAP@50 of 0.285 across Indian retail brands like Amul, Maggi, and Haldirams. - Developed a FastAPI backend for real-time inference and a React.js frontend supporting multi-image upload, confidence-based filtering, and automatic itemized bill generation with GST calculation and print export. - Addressed real-world ML challenges including class imbalance (avg. 3.3 samples/class), mixed annotation formats, and version compatibility across PyTorch and Ultralytics training environments.

Technical Skills

Languages: Python, Java, SQL

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Technologies/Frameworks: NumPy, pandas, Scikit-learn, Matplotlib, OpenCV, TensorFlow

Achievements

Certifications:

- Applied Machine Learning in Python (Coursera)
- Java Programming (HackerRank)
- Data Analytics Job Simulation (Deloitte Virtual Experience Program)

Competitive Programming:

- Solved **500+** algorithmic problems across **LeetCode**, **CodeForces**, and **GeeksforGeeks**.